

Smart Classroom Energy Saver.....

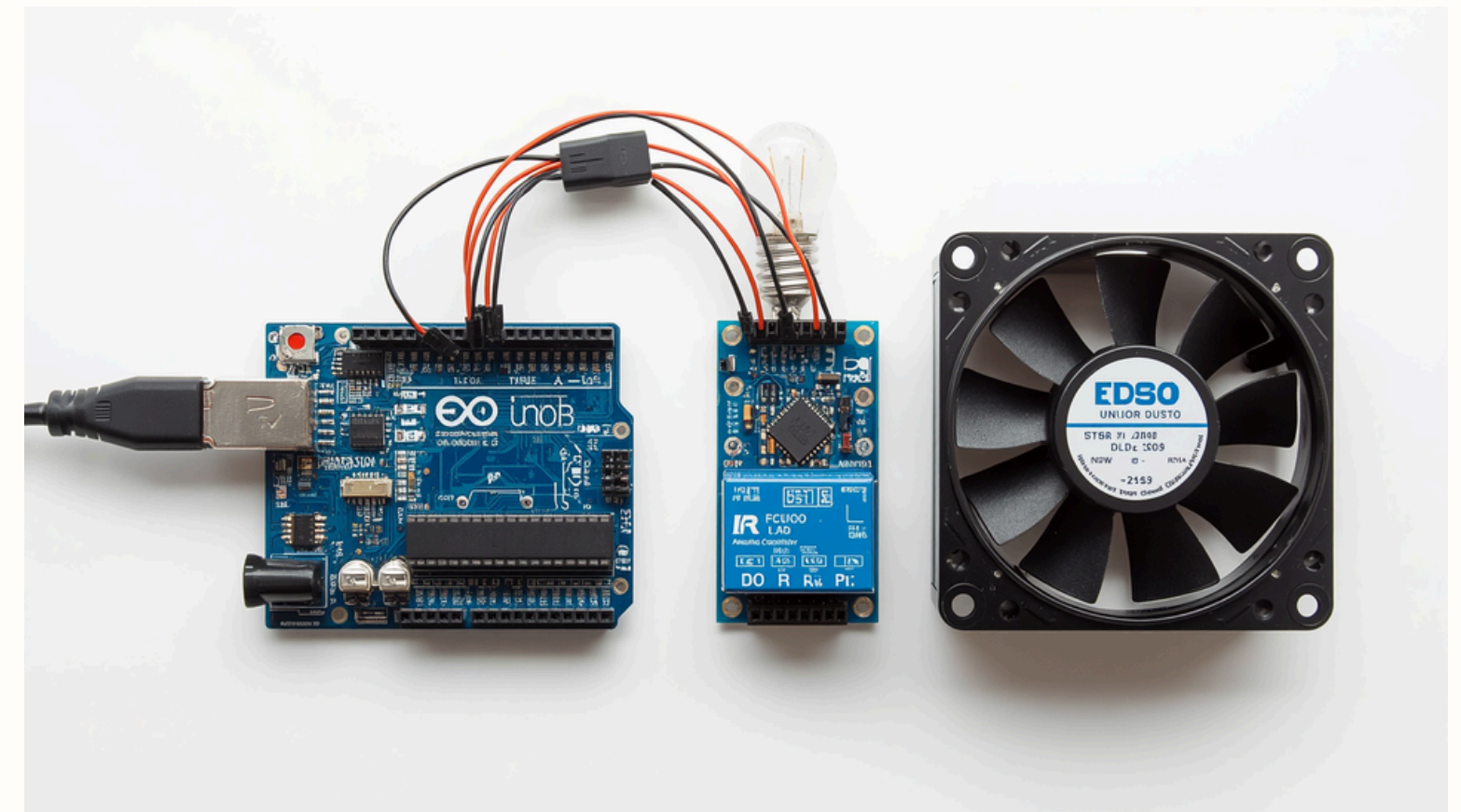
Arduino-Based Automated
Classroom Energy
Management System
Tech Fest 2026

The Problem

40-60%

Energy wasted in unoccupied classrooms

Classrooms often waste electricity when lights and fans remain ON even when empty. Studies show schools waste up to 30% of total energy budget annually due to unoccupied spaces left powered on.



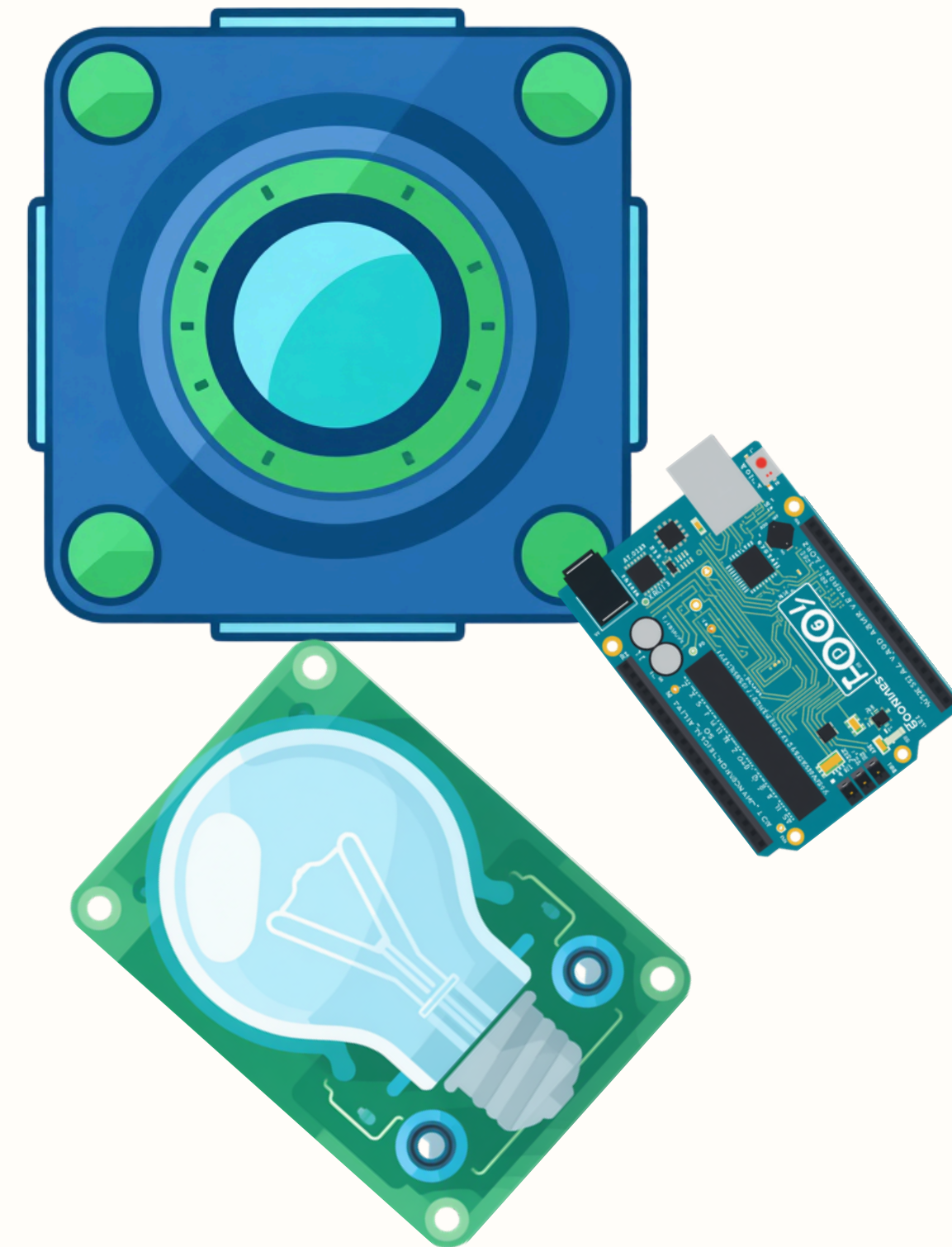
Project Objective

Develop an automated system that detects classroom occupancy and controls lights and fans to save electricity.

Workflow:

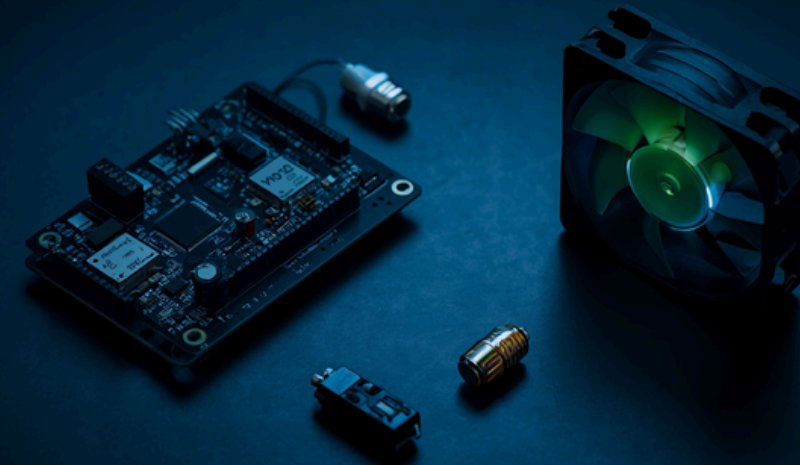
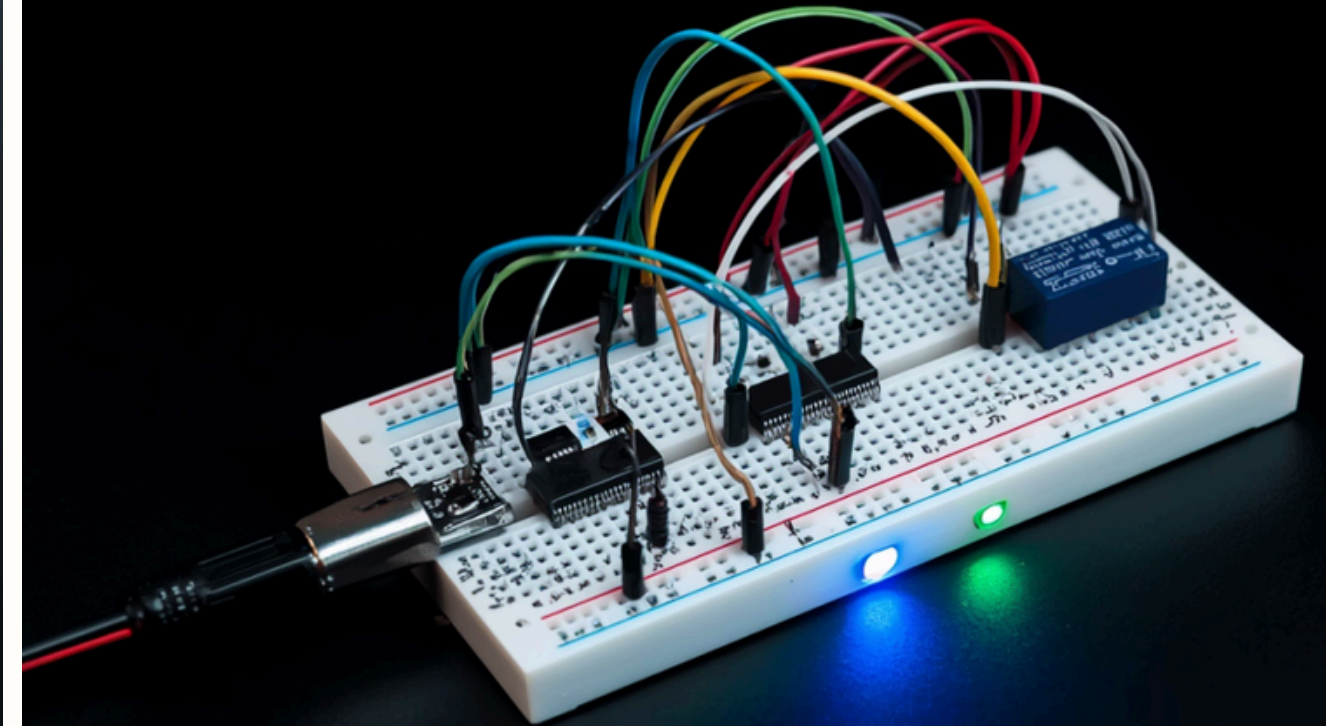
Sensor → Arduino → Relay → Control

- Ultrasonic & IR sensors detect presence
- Arduino processes occupancy data
- Relay module switches lights & fans
- Fully automated ON/OFF control

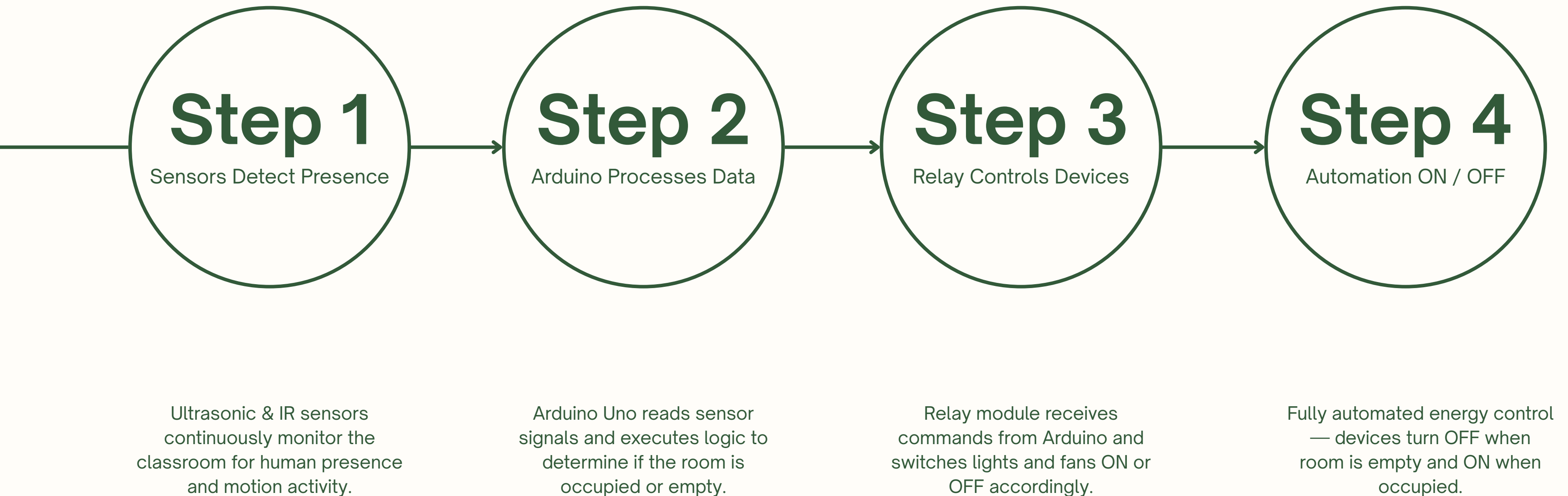


Components Used

Arduino Uno, Ultrasonic Sensor, IR Sensor, Relay Module, LED Bulb, DC Fan, Breadboard, Jumper Wires, Power Supply



How It Works



Advantages



Eco-Friendly

Saves Electricity & Reduces Bills

Automatically turns off lights and fans in empty classrooms, cutting unnecessary energy consumption and lowering electricity costs significantly.



Hands-Free

Automatic & Low-Cost Installation

Fully automated operation requires no manual intervention. Easy to install with affordable components, making it ideal for schools and institutions.

Applications



Schools & Colleges

Automated energy control in classrooms and lecture halls, reducing electricity waste during unoccupied periods and cutting school energy bills significantly.



Offices & Libraries

Smart occupancy detection in office spaces and library reading rooms ensures lights and fans operate only when people are present, maximizing efficiency.



Labs & Smart Buildings

Ideal for science labs, computer labs, and smart building infrastructure where automated control enhances both energy savings and operational convenience.

Future Scope

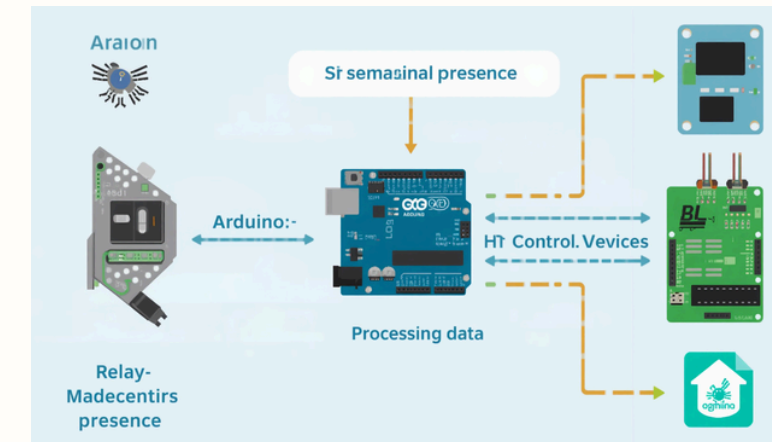
IoT-Based Remote Monitoring

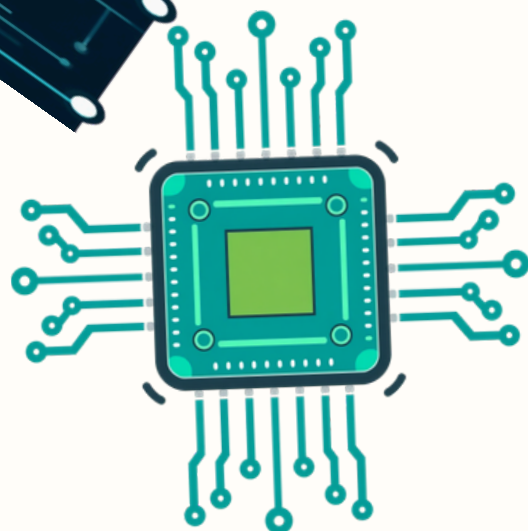
Mobile App Control

AI-Powered Occupancy Detection

Energy Usage Analytics Dashboard

Solar Power Integration





Thank You for Your Attention!
We welcome your questions & feedback
[✉ Project: Smart Classroom Energy Saver](#)
[🔗 Scan QR code for project details & code](#)

